

Cristian Palma - Researcher in Algorithms and Optimization

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Education:

PhD in Operations Research, Cornell University. Advised by Éva Tardos. *USA, 2024 - 2029*
MSc in Applied Mathematics, University of Chile. Advised by José A. Soto. *Chile, 2022 - 2023*
BSc in Mathematics and Computer Science Engineering, École CentraleSupélec. *France, 2019 - 2021*
BSc in Common Plan of Engineering, University of Chile. *Chile, 2017 - 2019*

Work Experience:

Operations Research Engineer at Center for Mathematical Modeling *Chile, 2024*
Applying Mixed-Integer Linear Programming techniques for solving a Crew Scheduling problem in Python.
Decreased computation time of crew schedules from a day to minutes by applying MILP techniques in Python.

Research Assistant at Center for Mathematical Modeling *Chile, 2022 - 2023*
Development of Approximation Algorithms for Concave Load Balancing.

Software Engineer Intern at Bigblue *France, 2021*
Problem solving and backend dev in Golang with FoundationDB on e-commerce logistics.

Teaching Assistantships:

Algorithmic Game Theory (2025) at Cornell University
Practical Tools for Operations Research, Machine Learning and Data Science (2025) at Cornell University
Graph Theory (2023) and **Calculability and Computational Complexity** (2023) at University of Chile
Mixed Linear Programming (2022, 2023) and **Combinatorial Algorithms** (2022) at University of Chile
Partial Differential Equations (2021), Measure Theory and Probability (2020) at École CentraleSupélec
Multivariable Calculus (2019) **Algorithms and Data Structures** (2018) at University of Chile
Differential and Integral Calculus (2023, 2022) at University of Chile
Introduction to Calculus (2023, 2018) and Algebra (2023, 2022) at University of Chile

Talks:

ORIE PhD Colloquium, Cornell University. Talk on [1]. *USA, 2025*
Young Researchers Workshop, Cornell University. Poster on [1]. *USA, 2025*
Algorithms, Game Theory, Combinatorics and Optimization (AGCO) Seminar. Talk on [1]. *Chile, 2025*
New York City Operations Day, New York University. Poster on [2]. *USA, 2025*
XIX Summer School in Discrete Mathematics, Center for Mathematical Modelling. Talk and Poster on [2]. *Chile, 2024*
Discrete Mathematics Session of the XCI Annual Meeting of the SOMACHI. Talk and Poster on [2]. *Chile, 2023*
Algorithms, Game Theory, Combinatorics and Optimization (AGCO) Seminar. Talk on [2]. *Chile, 2023*
Symposium of Graduate Students of the Engineering School, University of Chile. Talk and Poster on [2]. *Chile, 2023*

Publications:

- [1] Ariana Abel, Yoav Kolumbus, Jerónimo Martín Duque, Cristian Palma Foster, and Éva Tardos. Learning in strategic queuing systems with small buffers. *arXiv preprint arXiv:2502.08898*, 2025.
- [2] Cristian Palma Foster. Load balancing algorithms on scheduling problems with a concave function of the load. Master's thesis, Universidad de Chile, 2024. URL <https://repositorio.uchile.cl/handle/2250/200354>.

Honors:

Eiffel Excellence Scholarship granted by the Ministry of Europe and Foreign Affairs. *2019 - 2021*
First Place on Engineering Generation Ranking, University of Chile. *2018*
Andrés Bello Excellence Scholarship granted by the University of Chile. *2017 - 2022*
Golden Medal at the National Mathematical Olympiad of SOMACHI, Chile. *2016*

Projects:

Multipath Transmission Protocols for **Load Balancing on Networks** dev in Python. *2024*
Ranked 15th at IMC Challenge of Huawei.
Matching Algorithm for Language Tandem deployed at École CentraleSupélec *2020*
Matched 47% more people and reduced computation time to milliseconds with a Python implementation.

Computer Skills: Programming: Python, Matlab, Julia, Java, Clang, Golang and MySQL.

Languages: English (Fluent), French (Fluent), German (Elementary) and Spanish (Native).

Free Time: Volleyball, Bodybuilding, Dancing, Running, Kung Fu and **Competitive Programming**.